

Limits, Continuity and Differentiability

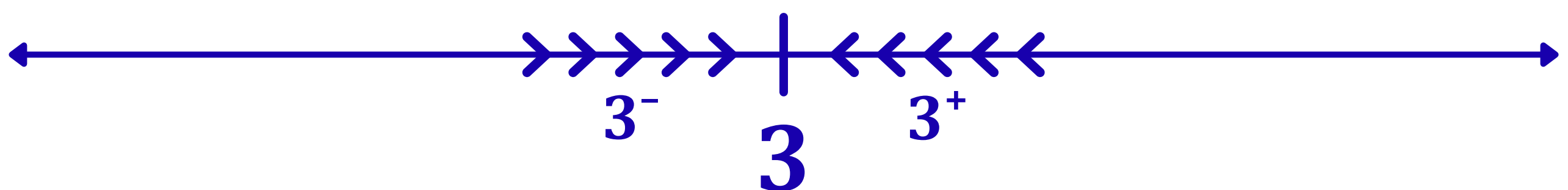
Introduction

$$f(x) = \frac{x^2 - 9}{x - 3}$$

This equation is **undefined at $x = 3$**

What value does **$f(x)$** approach when x approaches 3 (i.e., x gets very close to 3 but isn't exactly 3)? This can be found using limits.

$$f(3) = \frac{3^2 - 9}{3 - 3} = \frac{0}{0} = \textit{undefined}$$



$$f(x) = \frac{(x - 3)(x + 3)}{x - 3} = x + 3, x \neq 3$$

$f(x)$ is undefined at $x = 3$

But $f(x) = x + 3$ when $x \neq 3$

Left Hand Limit is the value of the function $f(x)$ at $x = a^- \rightarrow$ where a^- is slightly lesser than the value of $a \rightarrow$ For example $\rightarrow$

If $a = 3$

then $a^- = 2.99999999$

Right Hand Limit is the value of the function $f(x)$ at $x = a^+ \rightarrow$ where a^+ is slightly greater than the value of $a \rightarrow$ For example $\rightarrow$

If $a = 3$

then $a^+ = 3.0000001$

L.H.L of $f(x)$ at $x = a^-$ is denoted by $f(a^-)$

R.H.L of $f(x)$ at $x = a^+$ is denoted by $f(a^+)$

DEFINITION $\rightarrow$

The limit of the function $f(x)$ at $x = a$ exists when
LHL at $(x = a) = \text{RHL at } (x = a) = \text{Finite Quantity}$

Condition for limit to exist $\rightarrow$

LHL = RHL = Finite Quantity

$$LHL = \lim_{x \rightarrow a^-} f(x) = \lim_{h \rightarrow 0} f(a - h) = f(a^-)$$

$$RHL = \lim_{x \rightarrow a^+} f(x) = \lim_{h \rightarrow 0} f(a + h) = f(a^+)$$

Q: Check whether the limit for $f(x) = [\sin x]$ (where $[.]$ denotes GIF) exists at $x = 0$

$$LHL = \lim_{x \rightarrow 0^-} [\sin x] = \lim_{x \rightarrow 0^-} [-ve] = -1$$

$$RHL = \lim_{x \rightarrow 0^+} [\sin x] = \lim_{x \rightarrow 0^+} [+ve] = 0$$

$$LHL \neq RHL$$

Therefore, the limit for $f(x) = [\sin x]$ does not exist at $x = 0$

Q: Check whether the limit for the following function exists →

$$\lim_{x \rightarrow 2} \frac{|x - 2|}{x - 2}$$

$$LHL (-1) \neq RHL (1)$$

The limit does NOT exist.

Important Properties

$$\frac{1}{0^+} = \infty, \frac{1}{0^-} = -\infty$$

$$a^{-\infty} \rightarrow 0 \quad (a > 1)$$

$$a^{-\infty} \rightarrow \infty \quad (0 < a < 1)$$

Indeterminate Forms

$$\frac{0}{0}, \frac{\infty}{\infty}, \infty - \infty, 0 \times \infty, 1^\infty, \infty^0, 0^0$$

- The zeroes over here are NOT exact zeros, they are 0^+ and 0^- (tending to 0).
- The ∞ over here implies ∞^+ and ∞^- (tending to ∞).
- The 1 over here is NOT an exact 1. It is 1^+ and 1^- (tending to 1).

Algebra of Limits

$$\lim_{x \rightarrow a} g(x) = m$$

$$\lim_{x \rightarrow a} f(x) = l$$

$$\lim_{x \rightarrow a} [f(x) \pm g(x)] = l \pm m$$

$$\lim_{x \rightarrow a} [f(x) \times g(x)] = l \times m$$

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{l}{m}, m \neq 0$$

$$\lim_{x \rightarrow a} k[f(x)] = kl$$

Methods to Solve Indeterminate Forms

1. First Indeterminate Form

$$\frac{0}{0}$$

1. Factorization
2. Rationalization
3. Substitution
4. Some Standard Formula

(1) Factorization

$$\begin{aligned}\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} &= \lim_{x \rightarrow 2} \frac{(x - 2)(x + 2)}{(x - 2)} \\ &= \lim_{x \rightarrow 2} (x + 2) = 2 + 2 = 4\end{aligned}$$

(2) Rationalization

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{\sqrt{x + 4} - 2}{x} &= \lim_{x \rightarrow 0} \frac{\sqrt{x + 4} - 2}{x} \times \frac{\sqrt{x + 4} + 2}{\sqrt{x + 4} + 2} \\ &= \lim_{x \rightarrow 0} \frac{x + 4 - 4}{x(\sqrt{x + 4} + 2)} = \frac{1}{\sqrt{0 + 4} + 2} = \frac{1}{4}\end{aligned}$$

(3) L'Hopital Rule

L'Hopital's rule is a shortcut for evaluating limits of fractions that result in an "indeterminate form" like $\rightarrow$

$$\frac{0}{0}, \frac{\infty}{\infty}$$

Instead of getting a direct answer, you can find the limit by taking the derivative of the top function and the derivative of the bottom function separately, then re-evaluating the new fraction's limit. If the new limit is still indeterminate, you can repeat the process until the indeterminate form is gone and a finite quantity is obtained.

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{0}{0} \text{ or } \frac{\infty}{\infty}$$

$$= \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = \frac{0}{0} \text{ or } \frac{\infty}{\infty}$$

$$= \lim_{x \rightarrow a} \frac{f''(x)}{g''(x)} = \frac{0}{0} \text{ or } \frac{\infty}{\infty} \dots \text{continue}$$

Examples →

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{d}{dx}(1 - \cos x)}{\frac{d}{dx}(x^2)}$$

$$= \lim_{x \rightarrow 0} \frac{\sin x}{2x} = \frac{1}{2} \lim_{x \rightarrow 0} \frac{\sin x}{x} = \frac{1}{2}$$

$$\lim_{x \rightarrow 3} \frac{x^2 - 4x + 3}{x^3 - 2x^2 - 6x + 9}$$

$$= \lim_{x \rightarrow 3} \frac{\frac{d}{dx}(x^2 - 4x + 3)}{\frac{d}{dx}(x^3 - 2x^2 - 6x + 9)}$$

$$= \lim_{x \rightarrow 3} \frac{2x - 4}{3x^2 - 4x - 6} = \frac{2}{9}$$

Standard Limits

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{\sin^{-1} x}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{\tan^{-1} x}{x} = 1$$

$$\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$$

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log_e a$$

$$\lim_{x \rightarrow 0} \frac{\log_e (1 + x)}{x} = 1$$

2. Second Indeterminate Form

$$\frac{\infty}{\infty}$$

Methods for Solving this Indeterminate Form →

1. Take the highest power in the Numerator and the Denominator common from them separately.
2. L'Hopital Method Works for this Indeterminate form.
3. You can substitute this in place of $\infty \rightarrow$

$$x = \frac{1}{t}$$

Therefore:

When $x \rightarrow \infty$ then $t \rightarrow 0$

$$\lim_{x \rightarrow \infty} \frac{x^3 + x^2 + 2}{x^4 - x^2 + 1}$$

$$\lim_{x \rightarrow \infty} \frac{x^3 \left(1 + \frac{1}{x} + \frac{2}{x^2}\right)}{x^4 \left(1 + \frac{1}{x^2} + \frac{1}{x^4}\right)} = \lim_{x \rightarrow \infty} \frac{1}{x} = \frac{1}{\infty} = 0$$

When $x \rightarrow \infty$

$$\frac{1}{x} \rightarrow 0$$

3. Third Indeterminate Form

$$\infty - \infty$$

Methods for Solving this

Indeterminate Form $\rightarrow$

Convert it into these following

indeterminate forms $\rightarrow$

$$\frac{0}{0}, \frac{\infty}{\infty}$$

Example →

$$\lim_{x \rightarrow \infty} \left(\sqrt{x^2 - 8x} - x \right)$$

$$= \lim_{x \rightarrow \infty} \left(\sqrt{x^2 - 8x} - x \times \frac{\sqrt{x^2 - 8x} + x}{\sqrt{x^2 - 8x} + x} \right)$$

$$= \lim_{x \rightarrow \infty} \left(\frac{x^2 - 8x - x^2}{\sqrt{x^2 - 8x} + x} \right)$$

$$= \lim_{x \rightarrow \infty} \frac{-8x}{\sqrt{x^2 - 8x} + x} = \frac{\infty}{\infty}$$

Now that we converted this into another indeterminate form, we can solve this easily.

$$= \lim_{x \rightarrow \infty} \frac{x(-8)}{x \left(\sqrt{1 - \frac{8}{x}} + 1 \right)} = -4$$

4. Fourth Indeterminate Form

$$0 \times \infty$$

Methods for Solving this

Indeterminate Form →

Convert it into these other
indeterminate forms →

$$\frac{0}{0}, \frac{\infty}{\infty}$$

Example → where $[.]$ represents GIF

$$\lim_{x \rightarrow 0} \frac{x}{b} \left[\frac{a}{x} \right]$$

$$= \lim_{x \rightarrow 0} \frac{x}{b} \left[\frac{a}{x} - \left\{ \frac{a}{x} \right\} \right]$$

$$= \lim_{x \rightarrow 0} \left[\frac{a}{b} - \frac{x}{b} \left\{ \frac{a}{x} \right\} \right]$$

$$= \frac{a}{b}$$

5. Fifth Indeterminate Form

$$1^{\infty}$$

Methods for Solving this

Indeterminate Form $\rightarrow$

Use the given equation $\rightarrow$

$$\lim_{x \rightarrow a} [f(x)]^{g(x)} = 1^{\infty}$$

$$\lim f(x) = 1, \lim g(x) = \infty$$

$$\lim_{x \rightarrow a} [f(x)]^{g(x)} = e^{\lim_{x \rightarrow a} [f(x)-1][g(x)]}$$

$$\lim_{x \rightarrow 0} \left(\frac{3x^2 + 2}{7x^2 + 2} \right)^{\frac{1}{x^2}}$$

$$= e^{\lim_{x \rightarrow 0} \left(\frac{3x^2 + 2}{7x^2 + 2} - 1 \right) \frac{1}{x^2}}$$

$$= e^{\lim_{x \rightarrow 0} \left(\frac{3x^2 + 2 - 7x^2 - 2}{(7x^2 + 2)x^2} \right)}$$

$$= e^{\lim_{x \rightarrow 0} \left(\frac{-4x^2}{(7x^2 + 2)x^2} \right)}$$

$$= e^{-2}$$

6. Sixth Indeterminate Form

$$0^0, \infty^0$$

**Methods for Solving this
Indeterminate Form →**

Always take $\log_e$ on both sides

$$y = \lim_{x \rightarrow 0} x^x$$

$$\log_e y = \lim_{x \rightarrow 0} \log_e x^x$$

$$\log_e y = \lim_{x \rightarrow 0} x \log_e x$$

$$\log_e y = \lim_{x \rightarrow 0} \frac{\log_e x}{\frac{1}{x}} \rightarrow \frac{\infty}{\infty}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{d}{dx}(\log_e x)}{\frac{d}{dx}\left(\frac{1}{x}\right)} = \frac{\frac{1}{x}}{-\frac{1}{x^2}} = -x = 0$$

Sandwich Theorem

If there are 3 functions $f(x)$, $g(x)$, $h(x)$
such that $\rightarrow$

$$g(x) \leq f(x) \leq h(x)$$

If $\rightarrow$

$$\lim_{x \rightarrow c} g(x) = l$$

$$\lim_{x \rightarrow c} h(x) = l$$

Then $\rightarrow$

$$\lim_{x \rightarrow c} f(x) = l$$

Important GIF [.] Property

$$x - 1 < [x] \leq x$$

$$\lim_{x \rightarrow \infty} \frac{\{2x\}}{2x}$$

Using Sandwich Theorem

We know $\rightarrow 0 \leq \{2x\} < 1$

$$0 \leq \{2x\} < 1$$

$$\frac{0}{2x} \leq \frac{\{2x\}}{2x} < \frac{1}{2x}$$

$$\lim_{x \rightarrow \infty} \frac{0}{2x} \leq \lim_{x \rightarrow \infty} \frac{\{2x\}}{2x} < \lim_{x \rightarrow \infty} \frac{1}{2x}$$

$$0 \leq \lim_{x \rightarrow \infty} \frac{\{2x\}}{2x} < 0$$

**By sandwich theorem
we can say that $\rightarrow$**

$$\lim_{x \rightarrow \infty} \frac{\{2x\}}{2x} = 0$$

$$\lim_{x \rightarrow \infty} \frac{[x]}{2x}$$

Using Sandwich Theorem

We know $\rightarrow x - 1 \leq [x] < x$

$$x - 1 < [x] \leq x$$

$$\frac{x - 1}{2x} < \frac{[x]}{2x} \leq \frac{x}{2x}$$

$$\lim_{x \rightarrow \infty} \frac{x - 1}{2x} < \lim_{x \rightarrow \infty} \frac{[x]}{2x} \leq \lim_{x \rightarrow \infty} \frac{x}{2x}$$

$$\frac{1}{2} \leq \lim_{x \rightarrow \infty} \frac{[x]}{2x} < \frac{1}{2}$$

By sandwich theorem
we can say that $\rightarrow$

$$\lim_{x \rightarrow \infty} \frac{[x]}{2x} = \frac{1}{2}$$

$$\lim_{x \rightarrow 0} x^3 \sin \left(\frac{2}{x} \right)$$

Using Sandwich Theorem

We know $\rightarrow -1 \leq \sin a \leq 1$

$$-1 < \sin \left(\frac{2}{x} \right) \leq 1$$

$$-x^3 \leq x^3 \sin \left(\frac{2}{x} \right) \leq x^3$$

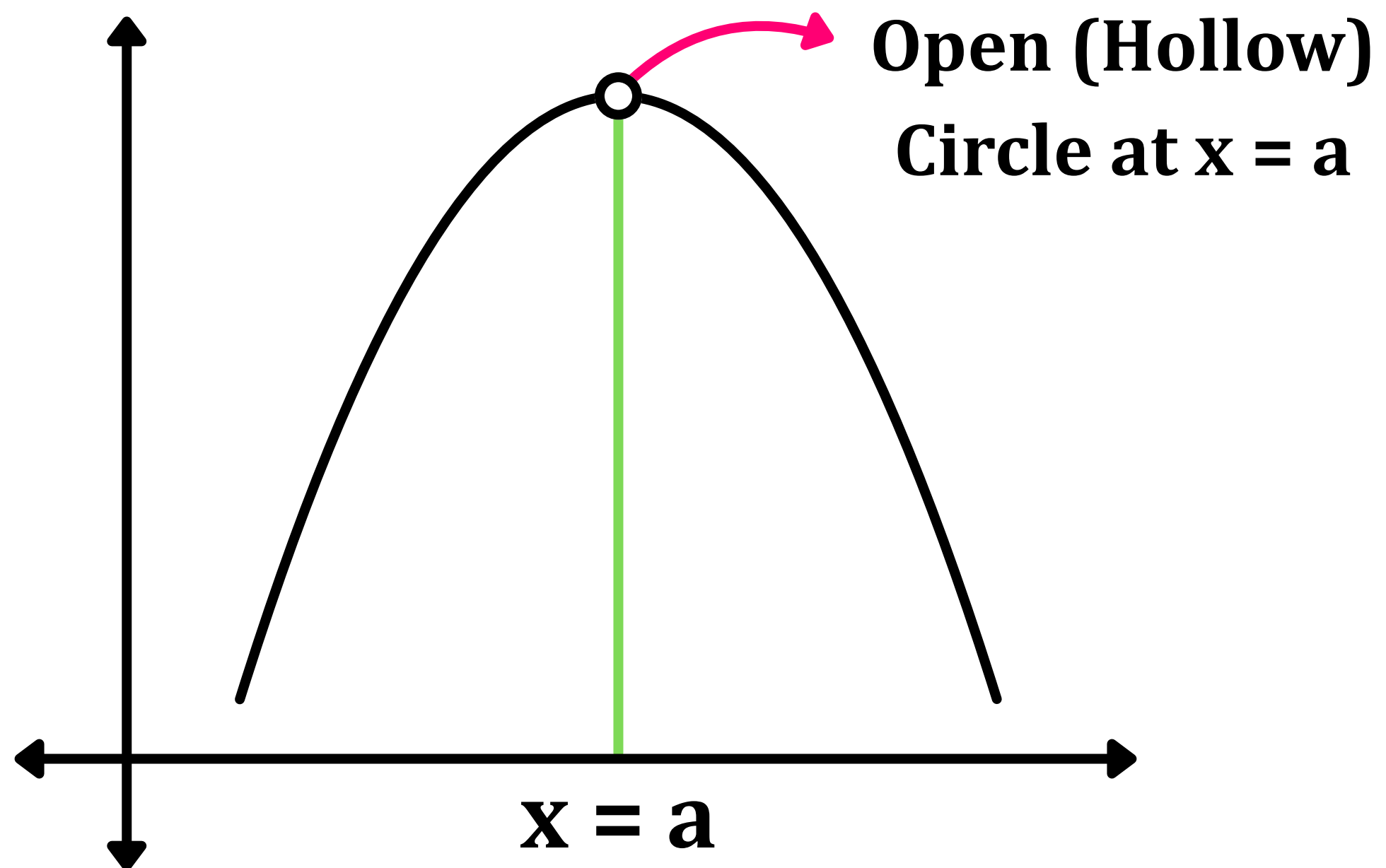
$$-\lim_{x \rightarrow 0} x^3 \leq \lim_{x \rightarrow 0} x^3 \sin \left(\frac{2}{x} \right) \leq \lim_{x \rightarrow 0} x^3$$

$$0 \leq \lim_{x \rightarrow 0} \sin \left(\frac{2}{x} \right) \leq 0$$

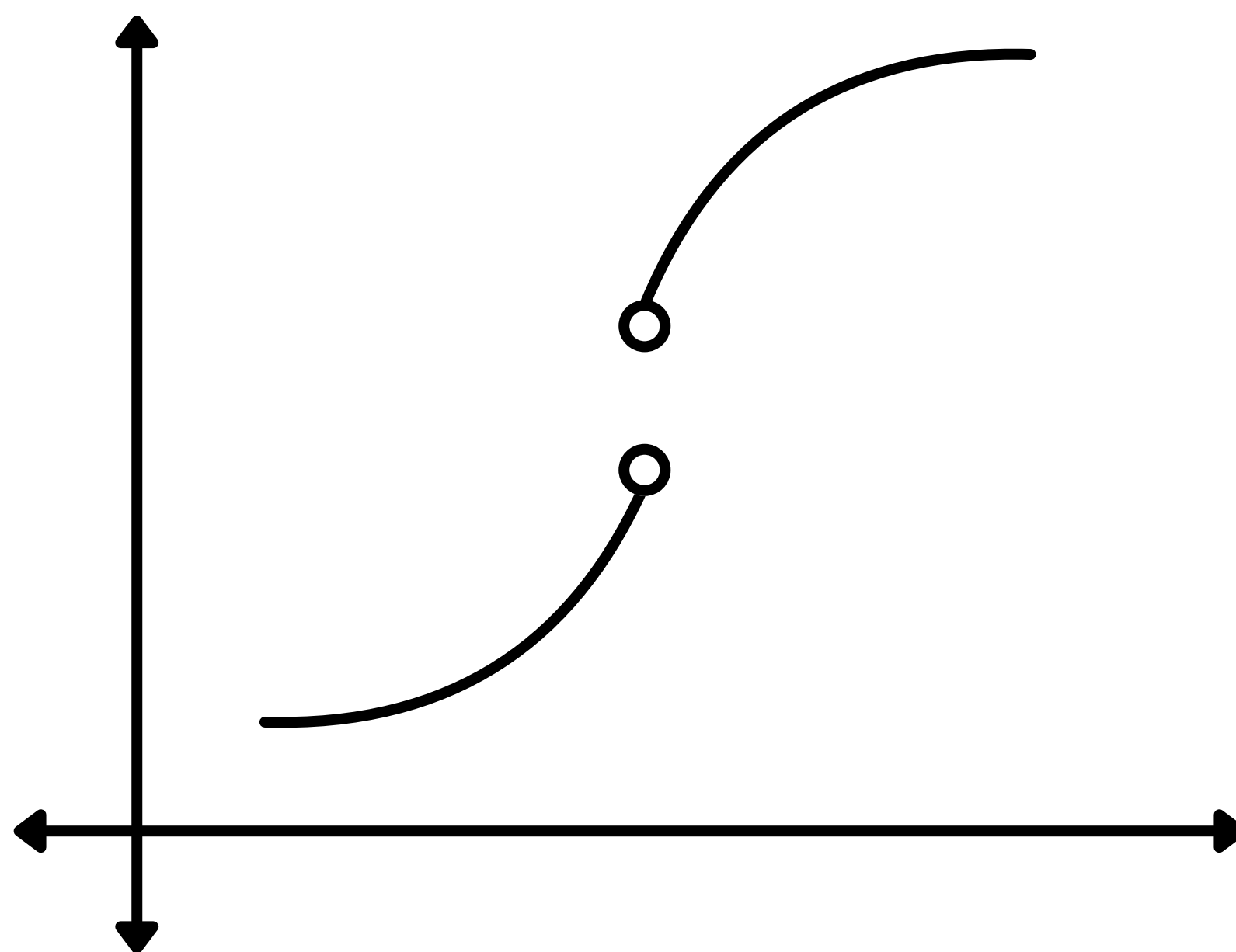
By sandwich theorem
we can say that $\rightarrow$ $\lim_{x \rightarrow 0} \sin \left(\frac{2}{x} \right) = 0$

Continuity of a Function

Graphically, a continuous function is a real valued function whose graph has no breaks.



Discontinuous Function at $x = a$



Discontinuous Function

**Condition for a function $f(x)$
to be continuous at $x = a \rightarrow$
 $LHL = RHL = f(a)$**

Generally, polynomial, logarithmic, trigonometric, exponential and modulus functions are continuous in their domain.

**Find whether the following function
 $f(x)$ is continuous at $x = 1$**

$$f(x) = \begin{cases} x + 1, & x \leq 1 \\ 2x - 1, & x > 1 \end{cases}$$

Let $x - 1 = h$ (so: $x = h + 1$)

Therefore, when $x \rightarrow 1$, $h \rightarrow 0$

$$LHL = f(1^-) = \lim_{x \rightarrow 1^-} (x + 1)$$

$$= \lim_{h \rightarrow 0} (1 + h + 1) = 1 + 1 = 2$$

$$RHL = f(1^+) = \lim_{x \rightarrow 1^+} (2x - 1)$$

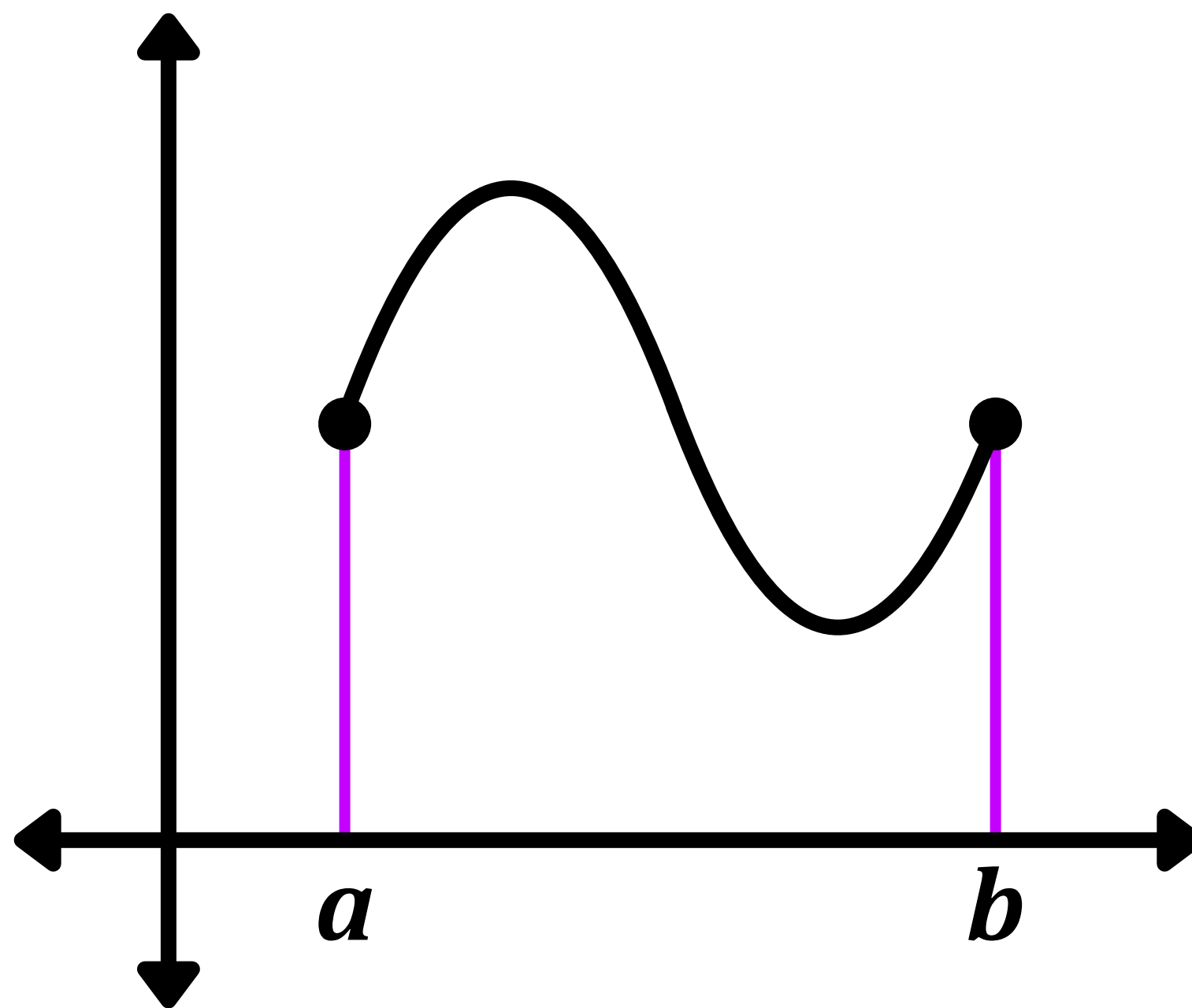
$$= \lim_{h \rightarrow 0} (2(h + 1) - 1) = 2 - 1 = 1$$

$$LHL(2) \neq RHL(1)$$

Hence the function $f(x)$ is NOT
continuous at $x = 1$

Continuity in an Interval

A function f is continuous in (a, b) if $\rightarrow$
 f is continuous at each and every point in
the interval (a, b) .



A function is continuous in $[a, b]$ if $\rightarrow$

(1) f is continuous in (a, b)

(2) f is right continuous at a , i.e. $\rightarrow$

$$RHL = \lim_{x \rightarrow a^+} f(x) = f(a) = \text{finite}$$

(3) f is left continuous at b , i.e. $\rightarrow$

$$LHL = \lim_{x \rightarrow b^-} f(x) = f(b) = \text{finite}$$



$\sin x$
 $\cos x$
 $\tan x$
 $\sec x$
 $\operatorname{cosec} x$
 $\cot x$
 $\ln x$
 e^x

Functions that are continuous in their domain

Modulus Function and Polynomial Function is always continuous.

$$y = [x] \rightarrow$$

Discontinuous when x is an integer

$$y = \{x\} \rightarrow$$

Discontinuous when x is an integer

$$y = \operatorname{sgn}(x) \rightarrow$$

Discontinuous when $x = 0$

For $[f(x)] \rightarrow$

Continuity should be checked at points when $f(x)$ becomes an integer.

For $\{f(x)\} \rightarrow$

Continuity should be checked at points when $f(x)$ becomes an integer.

For $\text{sgn}(f(x)) \rightarrow$

Continuity should be checked at points when $f(x)$ becomes 0.

Q: Comment on the continuity of the following function $f(x)$

$$f(x) \begin{cases} = 2x + 3, & -3 \leq x < -2 \\ = x + 1, & -2 \leq x < 0 \\ = x + 2, & 0 \leq x \leq 1 \end{cases}$$

**Points where continuity should be checked $\rightarrow$
 $\{-3, -2, 0, +1\}$**

At $x = 0 \rightarrow$

$$\begin{aligned} \lim_{x \rightarrow 0^-} f(x) &= \lim_{x \rightarrow 0^-} (x + 1) = \lim_{h \rightarrow 0} f(0 - h) \\ &= \lim_{h \rightarrow 0} (0 - h + 1) = 1 = f(0^-) \end{aligned}$$

$$\begin{aligned}\lim_{x \rightarrow 0^+} f(x) &= \lim_{x \rightarrow 0^+} (x + 2) = \lim_{h \rightarrow 0} f(0 + h) \\ &= \lim_{h \rightarrow 0} (0 + h + 2) = 2 = f(0^+)\end{aligned}$$

$$f(0^+) \neq f(0^-)$$

**Therefore, the equation is
discontinuous at $x = 0$**

At $x = -2 \rightarrow$

$$\begin{aligned}LHL &= \lim_{x \rightarrow -2^-} f(x) = \lim_{x \rightarrow -2^-} (2x + 3) \\ &= \lim_{h \rightarrow 0} f(-2 - h) \\ &= \lim_{h \rightarrow 0} (2(-2 - h) + 3) \\ &= \lim_{h \rightarrow 0} (-4 - 2h + 3) \\ &= \lim_{h \rightarrow 0} (-1 - 2h) = -1 = f(-2^-)\end{aligned}$$

$$\begin{aligned}RHL &= \lim_{x \rightarrow -2^+} f(x) = \lim_{x \rightarrow -2^+} (x + 1) \\ &= \lim_{h \rightarrow 0} f(-2 + h)\end{aligned}$$

$$= \lim_{h \rightarrow 0} (-2 + h + 1)$$

$$= \lim_{h \rightarrow 0} (-1 + h) = -1 = f(-2^+)$$

$$f(-2) = x + 1 = -2 + 1 = -1$$

$$f(-2^-) = f(-2^+) = f(-2) = -1$$

**Therefore, the equation is
continuous at $x = -2$**

At $x = -3 \rightarrow$

(Check whether it is right continuous)

$$RHL = \lim_{x \rightarrow -3^+} f(x) = \lim_{x \rightarrow -3^+} (2x + 3) = -3$$

$$f(-3) = 2x + 3 = 2(-3) + 3 = -3$$

**Therefore, the equation is
continuous at $x = -3$**

At $x = +1 \rightarrow$

(Check whether it is left continuous)

$$LHL = \lim_{x \rightarrow +1^-} f(x) = \lim_{x \rightarrow +1^-} (x + 2) = 3$$

$$f(+1) = x + 2 = 1 + 2 = 3$$

Therefore, the equation is
continuous at $x = +1$

Continuity of Composite Functions

To comment on the continuity of the
composite function: $f(g(x)) \rightarrow$

1. Check its continuity at the points where $g(x)$ can potentially be discontinuous.
2. If $f(x)$ is discontinuous when $x = a$, check the continuity of the composite function at the point where $g(x) = a$

$$E.g. \rightarrow f(x) = \frac{2}{2x-1}, g(x) = \frac{2+x}{x}$$

$$f(g(x)) = \frac{2}{2[g(x)]-1}$$

$$f(g(x)) = \frac{2}{2\left[\frac{2+x}{x}\right]-1}$$

$$x \neq 0$$

$$2\left[\frac{2+x}{x}\right]-1 \neq 0 \rightarrow 4+2x-x \neq 0$$

$$x \neq -4$$

Therefore, $f(g(x))$ is discontinuous when $\rightarrow$
 $x = \{-4, 0\}$

$$\lim_{n \rightarrow \infty} a^n = \begin{cases} 0 \rightarrow 0 < a < 1 \\ 1 \rightarrow a = 1 \\ \infty \rightarrow a > 1 \end{cases}$$

$$\begin{aligned} E.g. \rightarrow f(x) &= \lim_{n \rightarrow \infty} \frac{x^{2n} - 1}{x^{2n} + 1} \\ &= \lim_{n \rightarrow \infty} \frac{(x^2)^n - 1}{(x^2)^n + 1} \end{aligned}$$

$$(x^2)^n = a^n \rightarrow x^2 = a$$

Case (1) $\rightarrow 0 < a < 1 \rightarrow x^2 < 1$

$$f(x) = \lim_{n \rightarrow \infty} \frac{(x^2)^n - 1}{(x^2)^n + 1} = \frac{0 - 1}{0 + 1} = -1$$

Case (2) $\rightarrow a = 1 \rightarrow x^2 = 1$

$$f(x) = \lim_{n \rightarrow \infty} \frac{(x^2)^n - 1}{(x^2)^n + 1} = \frac{1 - 1}{1 + 1} = 0$$

Case (3) $\rightarrow a > 1 \rightarrow x^2 > 1$

$$f(x) = \lim_{n \rightarrow \infty} \frac{(x^2)^n - 1}{(x^2)^n + 1}$$

$$f(x) = \lim_{n \rightarrow \infty} \frac{(x^2)^n \left[1 - \frac{1}{(x^2)^n} \right]}{(x^2)^n \left[1 + \frac{1}{(x^2)^n} \right]} = 1$$

Therefore $\rightarrow$

$$\lim_{n \rightarrow \infty} f(x) = \begin{bmatrix} -1 \rightarrow x^2 < 1 \\ 0 \rightarrow x^2 = 1 \\ 1 \rightarrow x^2 > 1 \end{bmatrix}$$

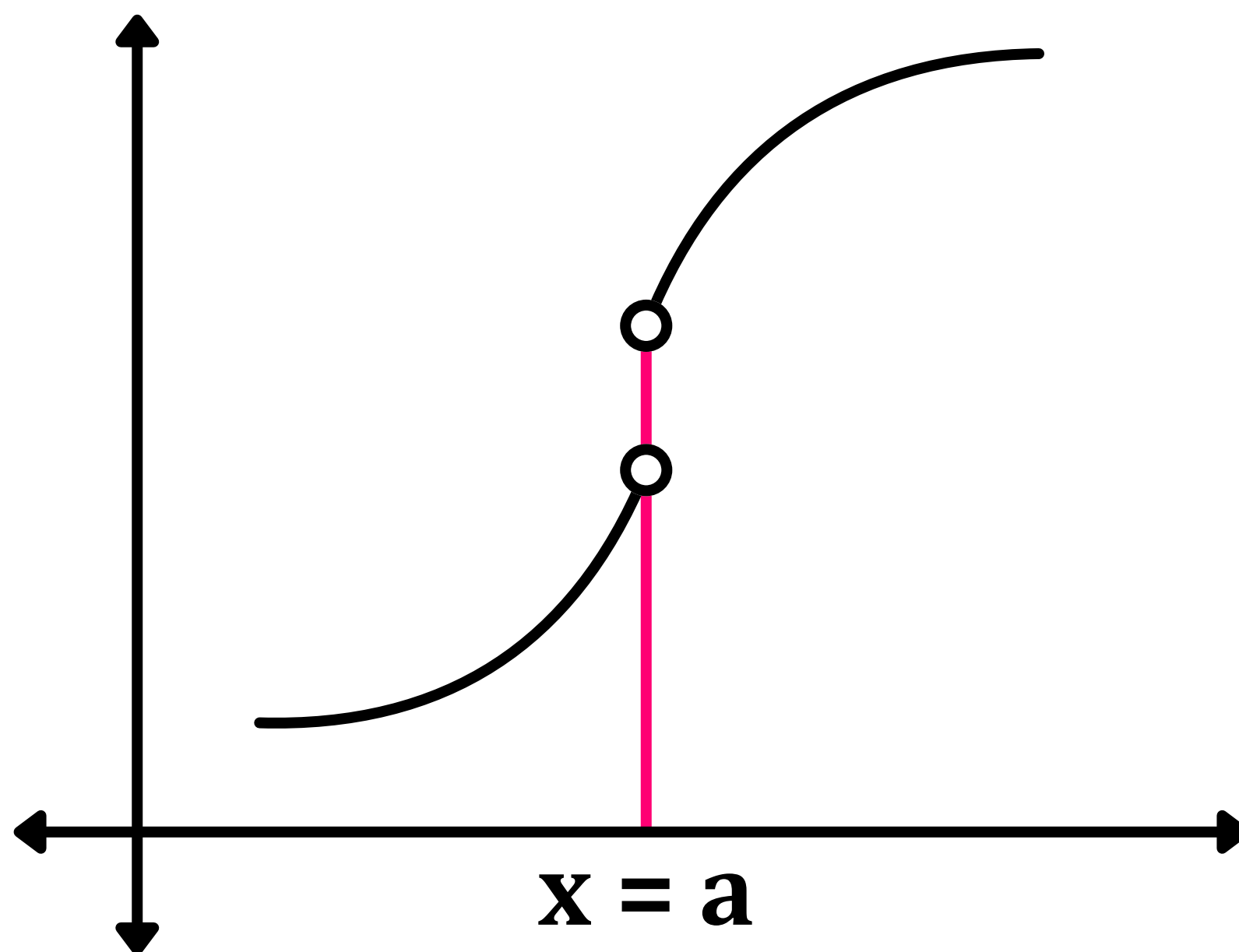
$$\lim_{n \rightarrow \infty} f(x) = \begin{bmatrix} -1 \rightarrow -1 < x < 1 \\ 0 \rightarrow x = \pm 1 \\ 1 \rightarrow x > 1 \text{ or } x < -1 \end{bmatrix}$$

**The given function is clearly
discontinuous when $x = 1$ or -1**

Differentiability and Condition for Differentiability

A function is said to be differentiable at point P if it has a unique tangent of finite length at point P.

A function is differentiable if it doesn't have a jump / break or a SHARP corner.

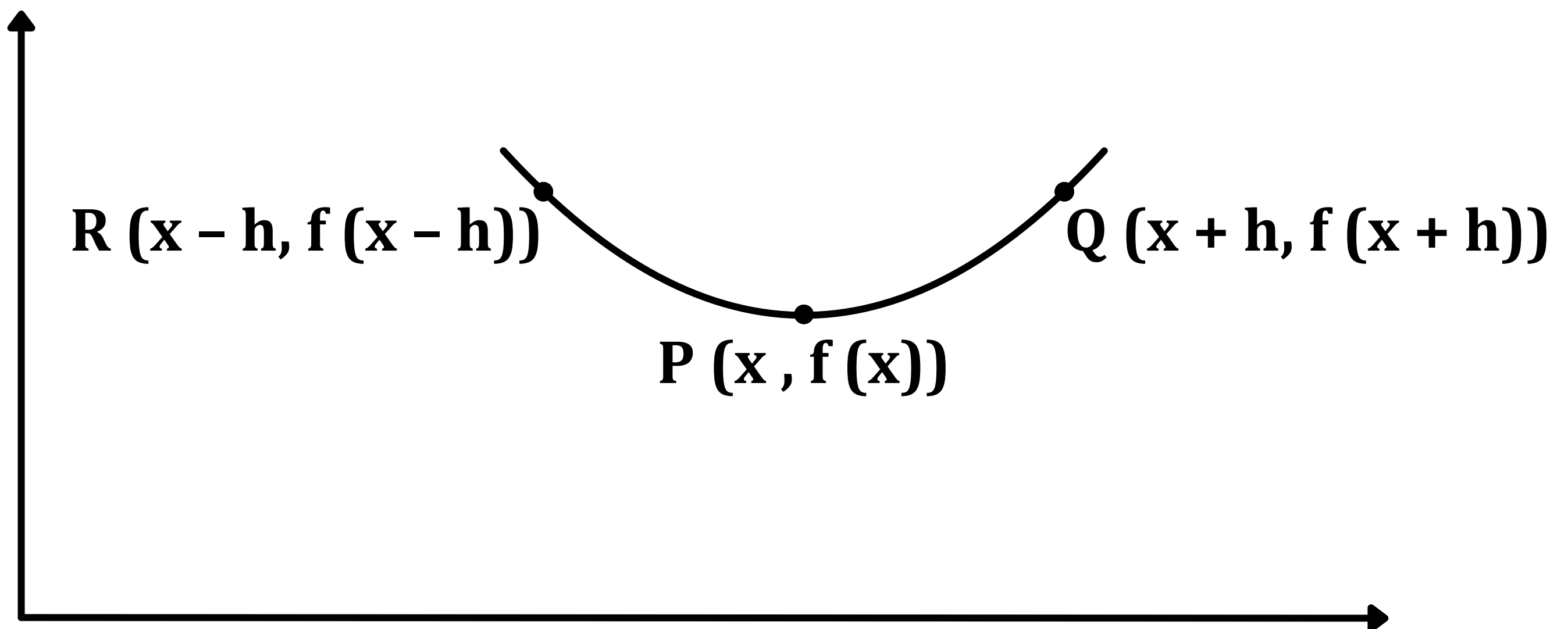


The given function is discontinuous at $x = a$

NOTE →

1. If a function is discontinuous at a point, it cannot be differentiable there
 2. But if a function is continuous at a point, it might or might not be differentiable.
- **Continuity is necessary for differentiability, but not sufficient.**

- E.g., $f(x) = |x|$ is continuous at $x = 0$ but not differentiable at $x = 0$ (sharp corner).



When $h \rightarrow 0$, line segments PR and PQ become tangents at point P

Initially $\rightarrow$

$$m_{PQ} = \frac{f(x+h) - f(x)}{h}$$

$$m_{PR} = \frac{f(x-h) - f(x)}{-h}$$

Since $h \rightarrow 0$

$$m_{PQ} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = RHD$$

$$m_{PR} = \lim_{h \rightarrow 0} \frac{f(x-h) - f(x)}{-h} = LHD$$

When $h \rightarrow 0$:

$m_{PR} = m_{PQ} = \text{Tangent at } P(x, f(x))$

$$\lim_{h \rightarrow 0} \frac{f(x-h) - f(x)}{-h} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Tangent at $P(x, f(x)) = \text{Derivative of } f(x) = f'(x)$

$$f'(x) =$$

$$\lim_{h \rightarrow 0} \frac{f(x-h) - f(x)}{-h} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \text{finite}$$

This is also called the **First Principle of Derivative**.

Q: Comment on the differentiability of the following function $f(x)$ at $x = 1 \rightarrow$

$$f(x) = \begin{cases} 2x, & x > 1 \\ x^2 - 1, & x \leq 1 \end{cases}$$

$$f'(1^-) = LHD = \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{(1-h)^2 - 1 - (1^2 - 1)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{h^2 - 2h}{-h} = \lim_{h \rightarrow 0} \frac{h - 2}{-1} = \lim_{h \rightarrow 0} 2 - h = 2$$

$$f'(1^+) = RHD = \lim_{h \rightarrow 0} \frac{f(1+h) - 0}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2(1+h) - 0}{h} = \lim_{h \rightarrow 0} \frac{2 + 2h}{h} = \frac{2}{0} = \infty$$

$$f'(1^+) \neq f'(1^-)$$

Hence, the given function is NOT derivable at $x = 1$

Q: Find the value of A and B if the following function $f(x)$ is differentiable at $x = 1 \rightarrow$

$$f(x) = \begin{cases} A + Bx^2, & x < 1 \\ 3Ax - B + 2, & x \geq 1 \end{cases}$$

$$f'(1^-) = LHD = \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{A + B(1-h)^2 - A - B(1)^2}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{A + B(1 + h^2 - 2h) - A - B}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{Bh^2 - 2Bh}{-h} = 2B - Bh = 2B$$

$$f'(1^+) = RHD = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{3A(1+h) - B + 2 - 3A(1) + B - 2}{h}$$

$$= \lim_{h \rightarrow 0} \frac{3Ah}{h} = 3A$$

$$f'(1^-) = f'(1^+) \Rightarrow 2B = 3A$$

Since $f(x)$ is derivable at $x = 1$, it is also continuous at $x = 1$.

$$f(1^-) = LHL = \lim_{x \rightarrow 1} f(x) = A + B(1)^2 = A + B$$

$$f(1^+) = RHL = \lim_{x \rightarrow 1} f(x) = 3A(1) - B + 2$$

$$= 3A - B + 2$$

$$f(1^-) = f(1^+) \rightarrow A + B = 3A - B + 2$$

$$2B = 2A + 2 \rightarrow B = A + 1$$

After solving the linear equations, we get $\rightarrow$

$$\mathbf{A = 2}$$

$$\mathbf{B = 3}$$

Main Reasons for Non-Differentiability:

1. Sharp Corner
2. Discontinuity
3. Vertical Tangents

When **LHD = RHD = ∞ or $-\infty$** $\rightarrow f(x)$ is **indifferentiable** but there is a **vertical tangent** at the given point.

Find the equation of tangent to the given function at $x = 0$.

$$f(x) = x^{\frac{1}{3}}$$

$$f'(x) = \frac{1}{3} x^{\frac{1}{3}-1} = \frac{1}{3} x^{\frac{-2}{3}} = \frac{1}{3x^{\frac{2}{3}}}$$

$$\lim_{x \rightarrow 0} f'(x) = \frac{1}{3x^{\frac{2}{3}}} = \frac{1}{0} = \infty$$

$$LHD = RHD = \infty$$

Only a vertical tangent is possible at $x = 0$ for the given function.